Samuel Shilling

+44 7713 915787 | shilling.sam@gmail.com | academic website

Profile

Quantitative researcher with expertise in developing Python-based data processing pipelines, modelling complex datasets using Bayesian inference, and building simulations to predict future results. A proven track record in two peer-reviewed first-author publications in a world-leading astrophysical research journal (a third in preparation), and presentations at five research conferences. Seeking to apply innovative scientific methods to solve challenging problems in industry.

Skills

- **Programming and tools:** Python, Linux, Git, NumPy, pandas, SciPy, Matplotlib, Ollama (local LLM integration), APScheduler.
- **Statistical analysis:** Data cleaning and visualisation, MCMC simulations, Spearman's rank and partial correlation tests, bootstrapping, Kolmogorov–Smirnov tests, bias tests.
- **Data modelling:** Bayesian inference, MCMC and nested sampling, posterior estimation, evidence-based model comparison, uncertainty quantification, non-linear least squares, linear regression, model extrapolation.
- **Communication and teamwork:** Presented to expert and non-expert audiences at five research conferences, published two scientific reports (third in progress), led teams on NASA and European Space Agency project proposals under short deadlines, assisted supervising four MSc student research projects.

Relevant experience and education

PhD in Astrophysics **Oct 2023 – Apr 2027 (expected)**
Lancaster University, UK

- Discovered a statistically significant correlation in a new data domain, ruled out sample and selection biases, and quantified its properties. Led to a publication.
- Developed a Python pipeline to model a large sample of complex astrophysical sources using Bayesian inference for model comparison and parameter estimation. Developed a simulation to test the inference. Publication in preparation.
- Presented research at five conferences (including three major national conferences) to expert and non-expert audiences.
- Assisted supervising four MSc Astrophysics student research projects, assisted with teaching undergraduate physics courses, and chaired the University Astrophysics journal club in rotation.
- Voluntarily developed a Python application (available on GitHub) that fully automates journal club role allocation indefinitely. This includes a locally hosted natural-language (Ollama) interface, so members can check and modify their roles in plain English, and a scheduled daily workflow.

Research Assistant in Astrophysics **Jun 2024 – Dec 2024**
NASA Goddard Space Flight Center, MD, USA

- Discovered data quality (rather than the physics) was a major driver of uncertainty in previous published results. Built a simulation to predict the results in future datasets. Led to a publication.
- Analysed newly detected transient astrophysical sources from NASA's *Swift* satellite, in real time (within minutes to hours) producing 37 community bulletin reports.

MPhys Physics and Astrophysics, First Class **Sep 2018 – Jun 2023**
University of Sheffield, UK

- Strong exam-based academic performance, achieving a first class degree with a final fourth-year average of 78%, and a dissertation mark of 80%.

First-author peer-reviewed publications

- Shilling et al. (in prep.) *Inferring GRB jet structures, physical parameters, and geometric properties from a large sample of X-ray afterglows.* In preparation (est. 2027).
- Shilling et al. (2026) *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* MNRAS, 552, stag1589
- Shilling et al. (2025) *Evidence for an intrinsic luminosity–decay correlation in GRB radio afterglows.* MNRAS, 542, 2421